

Column Generation for VRPTW LP Relaxation

Restricted Master Problem (RMP)

Minimize: $\sum_p c_p \lambda_p$

s.t. cover each customer ≥ 1

Solve LP $\Rightarrow$ dual prices π_i

Variables: route columns λ_p

dual prices π

Pricing Subproblem (ESPPRC)

Find route with $\bar{c} < 0$:

$$\bar{c} = c_p - \sum_i a_{ip} \pi_i$$

Shortest path with resource

constraints on $G = (V, A)$

new column

Repeat until no column with negative reduced cost exists $\Rightarrow$ LP optimal